SUMMER INTERNSHIP PROJECT REPORT

FINRELIEF AI

AI-POWERED DEBT RELIEF & FINANCIAL RECOVERY PLATFORM

A Project Report submitted in partial fulfillment of the requirements
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EXECUTIVE SUMMARY & ABSTRACT

In contemporary personal finance, outstanding debt accumulation represents a severe socioeconomic challenge for millions of individuals. Borrowers experiencing debt distress often face profound emotional stress, aggressive collection practices, and exorbitant advisory fees charged by private debt-settlement agencies. Although debt restructuring offers an accessible alternative to bankruptcy, traditional settlement solutions are opaque, costly, and financially predatory.

This summer internship project introduces FinRelief AI, an autonomous, enterprise-grade, full-stack financial recovery platform designed to democratize access to strategic debt relief and legal negotiation assistance. FinRelief provides users with immediate, data-driven financial diagnostics, objective health scoring, automated settlement lump-sum predictions, and customized creditor negotiation letters powered by a state-of-the-art dual-engine Generative AI architecture.

Key Engineering Innovations

- **Dual-Engine AI Orchestration:** Integrates Google Gemini LLMs (via the Google GenAI SDK) for empathetic financial roadmaps with a local deterministic fallback engine, guaranteeing 100% platform availability and <5ms failover during API disruptions or quota limits.
- **Comprehensive Financial Risk Analysis:** Calculates real-time monthly surplus, Debt-to-Income (DTI) ratio, EMI-to-income percentages, and an objective Financial Health Score (1–100) with dynamic risk level categorization.
- **Enterprise-Grade Security Architecture:** Built strictly aligned with OWASP Top 10 standards, featuring OAuth2 Password Bearer authentication, JWT session verification, RFC-6238 TOTP two-factor authentication (2FA), direct bcrypt salt hashing, and rate limiting.
- **Verified Quality Assurance & Zero Downtime:** Validated across 7 formal system specifications and 9 automated pytest modules achieving a 100% test pass rate with sub-50ms API endpoint latencies.

CHAPTER 1: INTRODUCTION & PROJECT OVERVIEW

1.1 Background & Societal Context

Personal debt in the modern economic landscape has surged exponentially across credit cards, unsecured personal loans, education debt, and medical expenses. A vast segment of middle- and lower-income earners find themselves in persistent debt traps characterized by high compound interest rates, negative monthly cash surpluses, and psychological anxiety. Commercial financial planners charge upward of $150 to $300 per consultation, effectively locking out the demographic that needs debt advisory services the most.

1.2 Problem Statement

Individuals burdened by debt suffer from three fundamental deficiencies: (1) lack of mathematical clarity regarding their true debt-to-income ratios and discretionary cash flow; (2) absence of objective negotiation benchmarks when interacting with predatory collection agencies; and (3) prohibitive financial barriers to obtaining professional hardship documentation.

1.3 Project Objectives

- **Democratize Financial Literacy:** Provide a 100% free, private web platform that empowers distressed borrowers without subscription barriers.
- **Automate Debt Health Diagnostics:** Calculate DTI, EMI burdens, and financial health scores instantaneously with strict cryptographic security.
- **Equip Users for Creditor Negotiations:** Predict realistic settlement targets (typically 40%–50% of outstanding balances) and generate formal legal hardship letters.
- **Achieve High-Availability AI:** Ensure zero-downtime AI advisory through automated circuit-breaker failover between cloud LLMs and rule-based fallback engines.

1.4 Scope of the Project

The project encompasses a decoupled full-stack architecture comprising a Python FastAPI REST backend, a responsive client frontend utilizing modern glassmorphism design, an encrypted SQLite relational database managed via SQLAlchemy ORM, and integrated Google Gemini generative capabilities.

CHAPTER 2: LITERATURE SURVEY & RELATED WORK

An extensive analysis of existing personal finance management (PFM) applications, commercial settlement companies, and AI advisory frameworks was conducted during the project ideation phase.

Platform / Solution	Core Capabilities	Identified Limitations	FinRelief AI Innovation
Traditional Debt Settlement Agencies	Human debt counselors, negotiation with creditors.	High upfront fees (15–25% of total debt), risk of fraud, slow communication.	100% free, autonomous, instant settlement predictions and legal letters.
Budgeting Apps (Mint, YNAB)	Expense categorization, bank account syncing.	No specialized debt settlement logic; no creditor negotiation letters or legal rights guidance.	Dedicated debt distress features: Snowball vs. Avalanche planners, FDCPA library, settlement targets.
Generic AI Chatbots (ChatGPT, Claude)	Open-ended conversational responses.	Lack financial math verification, hallucinate payment ratios, require manual prompt engineering.	Deterministic financial math engine coupled with verified, prompt-engineered hardship generation.

CHAPTER 3: REQUIREMENT ANALYSIS & TECHNICAL STACK

3.1 Functional Requirements

- **User Onboarding & Profile Management:** Secure user account creation, email verification token dispatch, and dynamic demographic updates.
- **Multi-Factor Authentication (2FA):** Two-factor authentication setup provisioning base32 TOTP secrets, QR codes, and time-window code validation.
- **Loan Portfolio CRUD:** Ingestion and tracking of multi-lender accounts with attributes including balance, APR, monthly EMI, and delinquency.
- **Mathematical Ratio Engine:** Real-time calculation of discretionary surplus, Debt-to-Income (DTI), and EMI-to-income percentages.
- **AI Debt Settlement Predictor:** Determination of optimal lump-sum settlement targets based on creditor risk matrices.
- **Automated Negotiation Letter Generator:** Contextual legal hardship letters customized by creditor name, account details, and tone.

3.2 Non-Functional & Security Requirements

- **Performance & Latency:** Core API endpoints must resolve in under 50 milliseconds; AI failover must execute in under 5 milliseconds.
- **Cryptographic Integrity:** Passwords hashed using bcrypt with dynamic salts; stateless session tokens signed via HS256 JWT.
- **Resiliency & Availability:** Zero downtime during external AI vendor rate limits (HTTP 429) or cloud downtime (HTTP 503).
- **Data Privacy:** Local database persistence ensuring zero external telemetry or third-party telemarketer sharing.

3.3 Technology Stack Summary

- **Backend:** Python 3.10+, FastAPI 0.111.0, Uvicorn 0.30.1, SQLAlchemy 2.0.31, Pydantic v2
- **Frontend:** HTML5, CSS3 (Glassmorphism, CSS Grid, Custom Animations), Vanilla JavaScript, Chart.js
- **Authentication:** OAuth2 Password Flow, python-jose (JWT), pyotp (RFC-6238 TOTP), bcrypt 4.1.3
- **AI & LLM Services:** Google GenAI SDK (Gemini 2.5/3.5 Flash), Deterministic Local Rule Fallback Engine
- **Quality Assurance:** Pytest 8.2.2, HTTPX 0.27.0, FastAPI TestClient, unittest.mock

CHAPTER 4: SYSTEM ARCHITECTURE & DESIGN

4.1 Decoupled Client-Server Architecture

FinRelief AI is engineered upon a high-performance decoupled architecture. The frontend operates as a responsive Single Page Application (SPA), communicating with the backend strictly via asynchronous REST API calls secured by Bearer JWT tokens. This separation ensures that client presentation and core financial computation remain completely modular.

4.2 Dual-Engine AI Orchestration & Fallback Circuit

To solve the critical reliability challenge of third-party LLM rate-limiting and service outages, FinRelief AI features an intelligent dual-engine architecture: When a user triggers an AI analysis or negotiation letter, the system dispatches the prompt to Google Gemini. If an APIError, 429 quota exhaustion, or 503 unavailable condition occurs, an automated circuit breaker immediately diverts execution to a local deterministic rule engine in <5ms. The user receives a fully structured, legally sound hardship letter without encountering application failure.

4.3 Database Schema & Entity Relational Design

The underlying relational database is architectured using SQLAlchemy ORM over SQLite, structured across 6 core entities:

- **users Table:** Stores UserID, Name, Email (unique index), Password (bcrypt hash), MonthlyIncome, MonthlyExpenses, VerificationToken, and TOTPSecret.
- **financial_profiles Table:** Maintains 1:1 relationship with users; stores calculated DTI_Ratio, EMI_Ratio, MonthlySurplus, and StressLevel.
- **loans Table:** Maintains 1:N relationship with users; tracks LenderName, LoanType, OutstandingAmount, InterestRate, EMI, and OverdueMonths.
- **ai_history Table:** Maintains historical audit trail of all AI-generated advice, prompt parameters, and timestamps.
- **settlement_predictions Table:** Stores predicted payout benchmarks, risk tiers, and target settlement amounts.
- **ai_negotiations Table:** Stores generated hardship negotiation letters and creditor settlement strategies.

CHAPTER 5: IMPLEMENTATION & CORE ALGORITHMS

5.1 Mathematical Financial Health Engine

The financial diagnostic subsystem evaluates borrower health using objective mathematical formulas rather than speculative approximations:

- **Monthly Cash Surplus:** Surplus = Monthly Income - (Monthly Living Expenses + Total Active Loan EMIs)
- **Debt-to-Income (DTI) Ratio:** DTI = (Total Monthly Debt Obligations / Gross Monthly Income) * 100
- **EMI-to-Income Burden:** EMI Ratio = (Total Monthly Loan EMIs / Gross Monthly Income) * 100
- **Financial Health Score (1–100):** Computed through a weighted algorithm penalizing high DTI (>40%) and delinquency while crediting positive cash surplus. Profiles are categorized into Low, Medium, High, or Critical Stress.

5.2 Strategic Debt Settlement Engine

The settlement prediction engine evaluates account delinquency and creditor risk to recommend strategic lump-sum settlement targets. For unsecured debts delinquent over 180 days, the engine calculates a recommended payout between 40% and 50% of the principal balance, enabling borrowers to negotiate realistically and avoid predatory terms.

5.3 Two-Factor Authentication (TOTP) Workflow

To guarantee enterprise-grade data security, the platform incorporates RFC-6238 TOTP two-factor authentication. Users configure 2FA using Google Authenticator via a generated base32 secret and QR code URI. On subsequent logins, a temporary intermediate JWT is issued until the user validates a dynamic 6-digit one-time password.

CHAPTER 6: QUALITY ASSURANCE, TESTING & VALIDATION

6.1 Formal System Test Matrix (TC-01 through TC-07)

Test ID	Test Category	Input Scenario	Expected Result	Obtained Result	Verdict
TC-01	Authentication	User signup with valid credentials & income	HTTP 200/201; JWT token; salted bcrypt password hash in DB	HTTP 200 OK; User created; Password encrypted	**PASSED**
TC-02	Authentication	Duplicate email registration	HTTP 400 Bad Request; prevent dirty database writes	HTTP 400; 'User with this email already exists'	**PASSED**
TC-03	Security	Access protected routes without Bearer JWT	HTTP 401 Unauthorized; challenge header present	HTTP 401 Unauthorized; route execution blocked	**PASSED**
TC-04	Financial Math	Income: $10k, Exp: $4k, EMI: $2k (Healthy)	Surplus: $4,000; EMI: 20%; Stress: Low; Score: 95.0	Surplus: $4,000.00; EMI: 20.00%; Score: 95.00	**PASSED**
TC-05	Financial Math	Income: $5k, Exp: $4k, EMI: $2k (Distressed)	Surplus: -$1,000; EMI: 40%; Stress: Critical; Score: 20.0	Surplus: -$1,000.00; EMI: 40.00%; Score: 20.00	**PASSED**
TC-06	Settlement Engine	CC Loan ($10k, 32% APR, 13 mo delinquent)	Target: 40%–50% (~$4,000 payout); Priority: Critical	Target: 40.00% ($4,000 target); Priority: Critical	**PASSED**
TC-07	AI Resilience	Simulated missing Gemini API key / timeout	Circuit-break to local fallback; is_fallback=True	Fallback executed in <5ms; full letter generated	**PASSED**

6.2 Automated Pytest Suite & CI/CD Mock Fixtures

To enable fast, zero-cost CI/CD verification without consuming external Gemini API quotas, an automated test harness was built using Pytest and unittest.mock. External LLM calls are mocked via @pytest.fixture, allowing the entire 9-module test suite to execute in under 0.5 seconds.

6.3 Critical Engineering Defects Remediated

- **Passlib / Modern Bcrypt Collision:** Eliminated deprecated passlib library; implemented direct bcrypt hashing (bcrypt.hashpw/checkpw) with dynamic salt generation.
- **Automatic Schema Initialization on Boot:** Configured Base.metadata.create_all directly inside the FastAPI startup event to eliminate 'no such table: users' exceptions.
- **SQLite Lock Contention in Synced Environments:** Resolved 'database is locked' errors in cloud-synced folders by tuning connection timeouts (timeout=30) and applying PRAGMAs journal_mode=DELETE and synchronous=NORMAL.

CHAPTER 7: RESULTS, PERFORMANCE & USER IMPACT

7.1 Empirical Performance Benchmarks

- **Core API Latency:** <50 milliseconds average response time across all CRUD and financial evaluation endpoints.
- **AI Failover Latency:** <5 milliseconds instantaneous failover from external cloud LLMs to the local deterministic rule engine.
- **Automated Test Pass Rate:** 100% verification across all unit, integration, and security test modules.
- **Zero Subscription Cost:** 100% free for end users, offering capabilities equivalent to commercial advisory firms charging $2,000+.

7.2 User Empowerment & Societal Impact

FinRelief AI transforms debt management from an overwhelming, predatory experience into an organized, mathematically clear recovery path. Borrowers gain actionable negotiation leverage, legal protection awareness through the built-in FDCPA library, and immediate relief from collection anxiety.

CHAPTER 8: CONCLUSION & FUTURE ENHANCEMENTS

8.1 Conclusion

The FinRelief AI summer internship project successfully designed, engineered, and verified a production-grade personal financial recovery web platform. By fusing modern full-stack web technologies (FastAPI, SQLAlchemy, responsive glassmorphism UI) with high-availability Generative AI orchestration, the platform proves that enterprise-quality financial advisory and debt settlement tools can be delivered autonomously, privately, and at zero cost.

8.2 Future Enhancements

- **Direct Open Banking Integration:** Incorporate secure Account Aggregator (AA) APIs to automatically synchronize live bank transactions and credit card balances.
- **PostgreSQL Cloud Migration:** Migrate from SQLite to PostgreSQL with read replicas to support enterprise-scale multi-tenant concurrency.
- **Native Mobile Applications:** Package the responsive web application into native iOS and Android apps using React Native or Flutter.

REFERENCES & TECHNICAL BIBLIOGRAPHY

[1] FastAPI Framework Documentation — High performance, easy to learn, fast to code, ready for production. https://fastapi.tiangolo.com/

[2] SQLAlchemy 2.0 Unified Documentation — Database Toolkit and ORM for Python. https://docs.sqlalchemy.org/

[3] Google Cloud Generative AI Python SDK Documentation — Google GenAI Developer Guide. https://ai.google.dev/

[4] Federal Trade Commission (FTC) — Fair Debt Collection Practices Act (FDCPA) Guidelines. 15 U.S.C. §§ 1692-1692p.

[5] OWASP Top Ten 2025: The Ten Most Critical Web Application Security Risks. Open Web Application Security Project. https://owasp.org/

[6] RFC 6238: TOTP — Time-Based One-Time Password Algorithm. Internet Engineering Task Force (IETF).

[7] Pytest: Modern Python testing tool with support for fixtures, assertions, and parameterized tests. https://docs.pytest.org/